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Coastal Change Detection of Henry & Lothian Island Using Multi-Year LULC Analysis and K-Means Clustering

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Coastal Change Detection of Henry and Lothian Island Using Multi-Year LULC Analysis and K-Means Clustering

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Abstract

The coast is one of the most dynamic and vulnerable terrestrial environments. Monitoring its change over time is very essential for sustainable management and planning. This paper proposes a systematic analysis of land use and land cover change detection in Henry and Lothian Islands, in the Indian Sundarbans delta, by using a multi-temporal satellite image and unsupervised machine learning. Ten yearly images (2014-2023) were processed for standardization and classification into major LULC classes: Water, Built-Up, Vegetation, and Bare Land. Data pre-processing was done by image normalization, resizing, denoising, removing outliers, and color-based rule mapping. K-Means clustering was applied on pixel trajectories to identify multi-year transition patterns showing regions with similar land cover evolution. The result shows that significant coastal transformation has taken place in built-up and vegetation categories, and some visible encroachment into water classes can also be observed. This study illustrates how effective unsupervised clustering can be at capturing complex coastal change dynamics and offers a promising analytical instrument for environmental monitoring.

Keywords—Coastal change detection, land use land cover (LULC), K-Means clustering, unsupervised learning, remote sensing, Sundarbans.

INTRODUCTION

Coastal areas undergo continuous change based on the natural activity of tides, erosion, deposition of sediment, and human development. The detection and quantification of such variations over successive years could deliver important information related to environmental health and human impact.

The Sundarbans is a UNESCO World Heritage Site, representing the world's largest contiguous mangrove forest system, spanning parts of India and Bangladesh. Henry and Lothian Islands, particularly, are sensitive areas within the delta that have experienced continuing shoreline changes and human modification of the land.

Active remote sensing and image analysis have become an integral part of LULC change assessment. Most conventional pixel-based classification techniques require labeled information; however, in many ecological studies, such datasets are not available. This limitation encourages the use of unsupervised learning methods such as K-Means clustering to extract intrinsic spatiotemporal patterns.

STUDY AREA

The given study pertains to Henry Island and Lothian Island, part of the southern Indian Sundarbans, ranging latitudinally from ~21°30′N to 21°45′N and longitudinally from 88°10′E to 88°30′E. The area is characterized by dense mangrove vegetation, tidal creeks,

mudflats, and settlements influenced by saline water intrusion and coastal erosion.

The islands represent an ideal case for monitoring coastal changes because they are ecologically sensitive and developmental activities like tourism and aquaculture are continuously pursued in these islands.

METHODOLOGY

It includes four major steps: data acquisition, preprocessing, classification, and change analysis.

A. Data Collection

Ten satellite-derived LULC images of 2014-2023 were collected and standardized. Preprocessing was carried out on every image using Python libraries (PIL, NumPy, SciPy) for maintaining consistent dimension, color, and noise level.

B. Data Preprocessing

All images were converted to RGB and brought into uniform dimensions. Images were enhanced using median filtering and unsharp masking to emphasize the boundaries between classes. Missing pixels and outliers were treated by interpolation and statistical replacement. Outliers greater than four standard deviations from the mean have been replaced by local medians in order to maintain spatial continuity and accuracy.

This preprocessing ensured that all the input data were homogeneous, denoised, and aligned spatially before any analysis.

C. LULC Classification

Each image was then transformed from color-coded representation to class IDs based on rule-based color mapping. The dominant land cover classes, which were determined, include:

Class 1: Water (Blue)

Class 2: Built-Up (Red)

Class 3: Vegetation (Green)

Class 4: Bare Land (Yellow)

Holes or trapped background regions were filled using morphological operations and the binary_fill_holes() function from SciPy, preserving continuous land areas and minimizing classification noise.

D. Multi-Year Change Detection

The change detection was carried out on the preprocessed stack of yearly classified images. A pixel-by-pixel comparison of the change

frequency between consecutive years allowed the identification of areas with the most active transformations during the decade.

E. K-Means Clustering for Pixel Trajectories

The unsupervised K-Means clustering algorithm was applied to identify recurring LULC transition patterns. Each pixel's multi-year class history formed a feature vector, and clustering revealed pixels with similar temporal behavior.

RESULTS AND DISCUSSION

The results obtained from the decadal LULC analysis provide broad insights into the coastal transformation dynamics of Henry and Lothian Islands. The pixel-based statistics clearly indicate changes in land composition, area proportion, and spatial transition between the primary land cover classes.

The dominance of water and vegetation during the early years of analysis, 2014–2016, signifies the natural mangrove and estuarine environment of the islands. Thereafter, the water area significantly decreased in the subsequent years, with many zones turning into either bare land or built-up areas. The pattern hints at both tidal channel shrinkage and anthropogenic occupation of intertidal zones.

The built-up area increased gradually and consistently throughout the decade. This expansion is most visible in land reclamation and tourism infrastructure near the northern and northeastern parts of Henry Island. The increase in the number of red-coded pixels across

time confirms progressive human settlement and developmental activity.

The vegetation cover varied significantly over the period. There was a temporary decline in the dense mangrove vegetation due to tidal intrusion and seasonal flooding in between 2016 and 2020 but partially recovered post-2020. The regrowth, however, did not match the earlier extent, hence signs of potential habitat loss.

Bare land, representing the exposed soil and sandbars, exhibited seasonality and tidal variation. Bare land pixels increased in years with poor vegetation and less precipitation, particularly near the edges and central patches of Lothian Island.

The computed transition frequency map revealed that the most change-prone areas are shoreline zones, transitional mangrove belts, and newly reclaimed lands. These are the zones which also reflect a high intensity of transformation, proving active coastal morphodynamics and pointing to the role of both natural and anthropogenic factors.

Results from trajectory clustering by K-Means gave a clear classification of the landscape into clusters according to similar patterns of temporal change. The stable clusters were regions that remained consistent in their LULC types throughout the decade, mainly comprising the dense mangrove areas and permanent water bodies. Dynamic clusters denoted regions with frequent changes between vegetation, bare land, and built-up categories. These dynamic clusters bring forward the areas prone to erosion and flooding, along with anthropogenic expansion. Overall, the analysis

underlines that Henry and Lothian Islands are undergoing progressive transformation marked by urban expansion, partial vegetation decline, and land reclamation in water-affected zones. A combination of statistical analysis, transition mapping, and unsupervised clustering provided a holistic understanding of the decadal-scale changes and their spatial distribution.

CONCLUSION

This study developed a robust and automated workflow to map multiyear coastal LULC changes using unsupervised K-Means clustering. The integration of preprocessing, classification, and clustering in the method assisted in identifying change hotspots and grouped regions based on temporal behavior.

The result shows that Henry and Lothian Islands are constantly changing, mainly by the expansion of built-up areas and decrease in vegetated cover. Tidal advance and habitat modification are exhibited in the coastal morphology. The proposed approach can be extended to other coastal ecosystems for long-term monitoring, policy formulation, and conservation planning.

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